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These solutions are not the only “perfect” or “10/10” solutions to be followed as a model. They are examples of ways to solve these problems. There are many many alternative ways to solve these problems. We hope that these example solutions help you check your work.

Problems

Q1. Let $S \subseteq V$ be a subset of a vector space. Recall that the span of S is:

$$\text{span}(S) = \{a_1\mathbf{v}_1 + a_2\mathbf{v}_2 + \cdots + a_n\mathbf{v}_n : a_i \in \mathbb{R} \text{ and } \mathbf{v}_i \in S\}.$$

Alternatively, $\text{span}(S)$ is the set of finite linear combinations of elements from S .

Prove the following:

(a) $S \subseteq \text{span}(S)$

Solution: Every vector $\mathbf{v} \in S$ can be written as $1 \cdot \mathbf{v}$, a linear combination of elements of S .

(b) If $S \subseteq T$ then $\text{span}(S) \subseteq \text{span}(T)$

Solution: We want to prove that every vector $\mathbf{v} \in \text{span}(S)$ is also an element of $\text{span}(T)$. To do this, start with a vector $\mathbf{v} \in \text{span}(S)$. By definition, there are vectors $\mathbf{v}_1, \dots, \mathbf{v}_n \in S$ and real numbers $a_1, \dots, a_n$ such that $\mathbf{v} = a_1\mathbf{v}_1 + \cdots + a_n\mathbf{v}_n$. Now, since $S \subseteq T$, we know that the vectors $\mathbf{v}_1, \dots, \mathbf{v}_n$ are also elements of T , and therefore $\mathbf{v}$ can be expressed as a linear combination of elements of T . Equivalently, $\mathbf{v} \in \text{span}(T)$, as desired.

(c) $\text{span}(S) = \text{span}(\text{span}(S))$

Solution: The fact that $\text{span}(S) \subseteq \text{span}(\text{span}(S))$ follows from (a); take $\text{span}(S)$ to be the set that you are spanning.

On the other hand, the inclusion $\text{span}(\text{span}(S)) \subseteq \text{span}(S)$ can be read in English as *a linear combination of linear combinations of elements of S , is a linear combination of elements of S* . So, let's start with some $\mathbf{v} \in \text{span}(\text{span}(S))$. This means that there exist some vectors $\mathbf{v}_1, \dots, \mathbf{v}_n \in \text{span}(S)$ and real numbers $a_1, \dots, a_n$ such that $\mathbf{v} = a_1\mathbf{v}_1 + \cdots + a_n\mathbf{v}_n$. Also, each one of the vectors $\mathbf{v}_i \in \text{span}(S)$, so there are vectors $\mathbf{w}_{i1}, \dots, \mathbf{w}_{im_i} \in S$ and real numbers $b_{i1}, \dots, b_{im_i}$ such that $\mathbf{v}_i = b_{i1}\mathbf{w}_{i1} + \cdots + b_{im_i}\mathbf{w}_{im_i}$. Expanding, we obtain that

$$\begin{aligned} \mathbf{v} &= a_1\mathbf{v}_1 + \cdots + a_n\mathbf{v}_n \\ &= a_1(b_{11}\mathbf{w}_{11} + \cdots + b_{1m_1}\mathbf{w}_{1m_1}) + \cdots + a_n(b_{n1}\mathbf{w}_{n1} + \cdots + b_{nm_n}\mathbf{w}_{nm_n}) \\ &= a_1b_{11}\mathbf{w}_{11} + \cdots + a_1b_{1m_1}\mathbf{w}_{1m_1} + \cdots + a_nb_{n1}\mathbf{w}_{n1} + \cdots + a_nb_{nm_n}\mathbf{w}_{nm_n}. \end{aligned}$$

Thus, $\mathbf{v}$ can be expressed as a linear combination of the $\mathbf{w}_{ij}$'s, which are all elements of S , and therefore $\mathbf{v} \in \text{span}(S)$.

Q2. Prove the following:

- (a) If $S \subseteq T$ and S is linearly dependent then T is linearly dependent.

Solution: If S is linearly dependent, there are some vectors $\mathbf{v}_1, \dots, \mathbf{v}_n \in S$ and real numbers $a_1, \dots, a_n$, **not all of them zero**, such that $a_1\mathbf{v}_1 + \dots + a_n\mathbf{v}_n = \mathbf{0}$. Since $S \subseteq T$, this is also a non-trivial linear combination of elements of T that results being $\mathbf{0}$. Thus, T is linearly dependent as well.

- (b) If $S \subseteq T$ and T is linearly independent then S is linearly independent.

Solution: Since T is linearly independent, for every linear combination

$$a_1\mathbf{v}_1 + \dots + a_n\mathbf{v}_n = \mathbf{0}$$

such that $\mathbf{v}_1, \dots, \mathbf{v}_n \in T$, we have that $a_1 = \dots = a_n = 0$.

Now, if we start with a linear combination

$$a_1\mathbf{v}_1 + \dots + a_n\mathbf{v}_n = \mathbf{0}$$

where $\mathbf{v}_1, \dots, \mathbf{v}_n \in S$, since $S \subseteq T$, $\mathbf{v}_1, \dots, \mathbf{v}_n \in T$. Therefore, $a_1 = \dots = a_n = 0$, implying that S is linearly independent.

Q3. A subset S' of S is proper if S' is not equal to S . We write $S' \subsetneq S$ to indicate that S' is a proper subset of S .

Prove: If S is linearly independent and $W = \text{span}(S)$ then no proper subset S' of S spans W .

Solution: Assume, looking for a contradiction, that there is a proper subset S' of S such that $\text{span}(S') = \text{span}(S)$ and take a vector $\mathbf{v}$ which is an element of S but not an element of S' . Since $\mathbf{v} \in S \subseteq \text{span}(S) = \text{span}(S')$, choose vectors $\mathbf{v}_1, \dots, \mathbf{v}_n \in S'$ and real numbers $a_1, \dots, a_n$ such that $\mathbf{v} = a_1\mathbf{v}_1 + \dots + a_n\mathbf{v}_n$. Equivalently, $a_1\mathbf{v}_1 + \dots + a_n\mathbf{v}_n - \mathbf{v} = \mathbf{0}$. But this is impossible (!), since this would give us a non-trivial linear combination of elements of S (to say, $\mathbf{v}, \mathbf{v}_1, \dots, \mathbf{v}_n$) that ends up being $\mathbf{0}$.

Q4. Suppose $S = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k\}$ is linear independent in V .

Prove that $T = \{\mathbf{v}_1, \mathbf{v}_1 + \mathbf{v}_2, \mathbf{v}_1 + \mathbf{v}_2 + \mathbf{v}_3, \dots, \mathbf{v}_1 + \dots + \mathbf{v}_n\}$ is linearly independent in V .

Solution: Lets start with a linear combination of T

$$a_1 \mathbf{v}_1 + a_2 (\mathbf{v}_1 + \mathbf{v}_2) + \dots + a_n (\mathbf{v}_1 + \dots + \mathbf{v}_n) = \mathbf{0},$$

and just note the the linear combination above can be rewritten as

$$(a_1 + \dots + a_n) \mathbf{v}_1 + (a_2 + \dots + a_n) \mathbf{v}_2 + \dots + a_n \mathbf{v}_n = \mathbf{0}.$$

Now, since S is linearly independent, then

$$\begin{aligned} a_n &= 0 \\ a_{n-1} + a_n &= 0 \\ &\vdots \\ a_2 + \dots + a_n &= 0 \\ a_1 + a_2 + \dots + a_n &= 0 \end{aligned}$$

which implies that $a_1 = a_2 = \dots = a_n = 0$. Since we started with an arbitrary linear combination of T that equals $\mathbf{0}$, and we concluded that the coefficients in such a combination have to be all 0, we can conclude as well that T is linearly independent.

Q5. Let $W \subseteq V$ be a subset of a vector space. Prove: W is a subspace if and only if $W = \text{span}(W)$.

Solution: For the forward implication let's assume that W is a subspace of V and let's conclude that $W = \text{span}(W)$. By Q1(a), $W \subseteq \text{span}(W)$, so we only need to check that $\text{span}(W) \subseteq W$. To do this, let $\mathbf{v} \in \text{span}(W)$, that is, there are vectors $\mathbf{v}_1, \dots, \mathbf{v}_n \in W$ and real numbers $a_1, \dots, a_n$ such that $\mathbf{v} = a_1 \mathbf{v}_1 + \dots + a_n \mathbf{v}_n$. But W is a vector subspace of V (!) so it is closed under linear combinations. Since all the $\mathbf{v}_i$'s are in W , $\mathbf{v}$ is in W as well.

For the other implication, if W can be expressed as a span (in particular, its own span), then it has to be a vector subspace of V , since the span of any subset of V is indeed a vector subspace of V .

Q6. Prove $\text{span}(\{1, 1+x, 1+x+x^2\}) = \mathbf{P}_2(\mathbb{R})$

Solution: Let $p(x) = a_0 + a_1x + a_2x^2 \in \mathbf{P}_2(\mathbb{R})$ be arbitrary. To prove that

$$p(x) \in \text{span}\{1, 1+x, 1+x+x^2\},$$

we just need to express $p(x)$ as a linear combination of 1, $1+x$, and $1+x+x^2$. Note that

$$p(x) = a_0 + a_1x + a_2x^2 = (a_0 - a_1) \cdot 1 + (a_1 - a_2)(1+x) + a_2(1+x+x^2).$$

Q7. Suppose $\{\mathbf{u}, \mathbf{v}, \mathbf{w}\}$ is linearly independent in V .

Prove that $\{\mathbf{u} + \mathbf{v}, \mathbf{u} + \mathbf{w}, \mathbf{v} + \mathbf{w}\}$ is also linearly independent in V .

Solution: As usual, let's start with a linear combination

$$a(\mathbf{u} + \mathbf{v}) + b(\mathbf{u} + \mathbf{w}) + c(\mathbf{v} + \mathbf{w}) = \mathbf{0}$$

and note that this linear combination can be rewritten as

$$(a + b)\mathbf{u} + (a + c)\mathbf{v} + (b + c)\mathbf{w} = \mathbf{0}.$$

Now, since the set $\{\mathbf{u}, \mathbf{v}, \mathbf{w}\}$ is linearly independent, we obtain that

$$a + b = 0$$

$$a + c = 0$$

$$b + c = 0$$

which implies that $a = b = c = 0$.